

Economy Action Plan — every item, every security fix

HIGH ROLEPLAY · 16 July 2026 · Design chosen by you: **no craft time, no cooldowns** — prices alone must make every NPC circle unprofitable. Crafter profit comes from selling to **players**.

The 3 rules every price below follows

Rule 1 — No profitable NPC circle. If all materials can be bought from NPCs, the NPC must pay **10–25% less** than the material cost. (Margin must be clearly negative, because clicking is free now.)

Rule 2 — Profit = gathering, risk, or players. NPCs pay real profit only for things that cannot be bought: mined ore, grown leaves, robbery loot, junk from bins. Crafters (medics, gangs, cooks) earn by selling to **players** in the window: *craft cost < player price < NPC shop price*.

Rule 3 — The server decides everything. Price and recipe always come from the server config, never from the client.

A. Security fixes (code) — do these first

#	Priority	What to do	Where
1	CRITICAL	Stop paying the client's price. In the sell handler, ignore the <code>price</code> argument. Build one server-side price map from the config (mining sell list, drug dealer, weapon dealer, EMS sell list, junk list). If the item is not in the map → refuse the sale. This kills the "sell 1 water for 999,999,999" hack.	<code>features/shop-manager-items/SellService.ts</code>
2	CRITICAL	Stop trusting the client's recipe. For every craft type (mining, ingot, ems, gang drug), look the recipe up in the server config and take the materials the recipe says — exactly like the gang weapon bench already does. Cap amount per request (e.g. 20).	<code>craft/craftItem.ts</code>
3	HIGH	Delete the dead timer code (you chose no craft time — remove it so nobody half-revives it): the crafting DB model, collect handler, status handler, reduce-time handler, and the unused second craft path. Also delete the never-read <code>limitation</code> / <code>sellLimitation</code> fields from the EMS config and the <code>craftTimeMinutes</code> fields from all recipes.	<code>models/crafting_model.ts</code> , <code>craft/collectItems.ts</code> , <code>craft/getCraftStatus.ts</code> , <code>reduceCraftingTime</code> , <code>features/craft-system/*</code> , <code>SHOPS/EMS.js</code>
4	HIGH	Log every sale and craft (player, item, amount, money, time) and make a daily "top 10 earners / top items sold" report. With no cooldowns, monitoring is your alarm system.	server (new small service)
5	HIGH	Add a circle-checker to the generator. A script inside <code>npm run generate</code> that computes the NPC-material cost of every recipe and fails the build if any NPC pays more than that cost. This makes Rule 1 automatic forever.	<code>generate_economy_config.js</code>
6	HIGH	Port the military-component recipes into the generator (knife, machete, lock pick, all guns, vests, ammo — copy from the deployed config), delete the old iron-bar/part recipes, then regenerate and copy to both high-ui and high-roleplay-v1. Do this before any regenerate, or the live recipes get wiped.	<code>CRAFTS/GANG.js</code>

7	RULE	Never add an NPC buyback for: ammo, vests, cooked food, reinforced steel, or any item whose materials are 100% NPC-bought. No buyback = no circle possible.	design rule
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B. Mining shop — every item

The fix here is on the **buy** side: raise ore prices so "buy ore → refine → sell bars" loses money, while **mined** (free) ore keeps full refining profit. Bar sell prices stay, so miners lose nothing.

Item	How obtained	Shop sells at (buy)	Shop pays (sell)	NEW value	Action	Why
Stone ore	mined / shop	22	16	—	KEEP	Sell < buy, no circle.
Sulfur ore	mined / shop	30	24	—	KEEP	Sell < buy. (Also ammo material — ammo has no buyback.)
Coal ore	mined / shop	35	26	—	KEEP	Sell < buy.
Iron ore	mined / shop	37	26	buy: 65	CHANGE	Kills printer #1: 100 bought ore + coal = 6,850 > 6,048 bar revenue (−12%). Mined ore still earns ~5,700 per refine.
Gold ore	mined / shop	53	37	buy: 85	CHANGE	Bought-ore gold bars: cost 11,450 vs revenue 8,640 (−25%). Mined gold stays profitable.
Iron bar	refined	2,100	1,512	—	KEEP	Safe once ore costs 65. Buy-bar→sell-bar already a loss.
Gold bar	refined	3,300	2,160	—	KEEP	Safe once gold ore costs 85.
Reinforced steel	refined / shop	5,900	no buyback	—	KEEP	No buyback = no circle. Never add one (Rule A7).
Gilded iron	refined (needs 2 gems)	—	21,600	—	KEEP	Gems are 0.06% mining drops — gathering-gated. Selling raw gems pays slightly more, so no printer.
Auriron	refined (needs 4 gems)	—	30,240	—	KEEP	Same — gem-gated, refining is a small loss vs raw gems.
Amberite / Aqualith / Luminite gem	mined (0.06%)	—	86,400	—	KEEP	Pure gathering jackpot. Fine.
Pickaxe lvl 1–4	crafted / shop / premium	7,200 (lvl1)	no buyback	—	KEEP	No buyback; crafting from bought mats costs more than the shop price → no circle.
Anti-sharp / Gas / Radioactive kit	crafted / shop	22,000 / 70,000 / 110,000	no buyback	—	KEEP	Same: bought-material craft cost (≈38,700 / 83,800 / 139,500) is above shop price; no buyback.
Mining map	supermarket 3,000	3,000	no buyback	—	KEEP	Consumable, buy-only.

C. Drug dealer — every item

Principle: leaves and weed are **grown** → NPC pays profit. Chemicals from the supermarket are **bought** → any product made only from them must sell at a loss. Ladder must always point up: product > its parts.

Item	Materials (per unit)	Cost/unit	Dealer pays now	NEW	Action	Why
Weed	grown	time	299	—	KEEP	Gathering income.
Coca leaves	grown	time	356	—	KEEP	Gathering income.
Saffarole leaves	grown	time	402	—	KEEP	Gathering income.
Saffarole oil	1 leaf (grown)	402*	862	—	KEEP	All-gathered chain; +460 rewards the grower. No NPC circle possible.
Sulfuric acid	2 batteries (NPC)	1,000	no buyback	—	KEEP	Intermediate only — never add a buyback.
Acetone	1 paint color (NPC)	600	no buyback	—	KEEP	Same. (EMS shop sells it at 2,000 — buy side only, fine.)
Ammonia	2 house cleaner (NPC)	1,200	no buyback	—	KEEP	Same.
Methylamine	3 methanol + 2 ammonia — all NPC	4,800	6,554	4,000	CHANGE	Kills printer #2 (was +1,754/click, ~630k/hr). New: −17%. Still valuable as MDMA ingredient.
MDP2P	acetone + saff. oil + ammonia	2,662*	5,979	3,500	CHANGE	Was +172%. New: +838 per unit — pays the leaf grower, not the clicker.
Cocaine powder	3 acetone + 3 leaves + 2 acid + 2 bicarb → 3	1,989*	4,830	3,800	CHANGE	Was +143%. New: profit ≈ leaf-growing wage. Needs grown leaves, so no pure-NPC circle.
Cocaine pack	1 powder + 1 plastic pack (433 NPC)	4,233*	5,000	4,600	CHANGE	Fixes the inverted ladder: with powder at 3,800 the pack is finally worth more than its parts (+8.7%).
MDMA powder	2 mdp2p + 2 methylamine → 3	5,000*	4,485	5,500	CHANGE	Today it sells below its own ingredients (dead recipe). New: +10% over parts.
MDMA pill	1 powder + 1 capsule (250 NPC)	5,750*	10,925	6,200	CHANGE	Kills printer #3 (was +122%; a 250\$ capsule multiplied value 2.4×). New: +7.8%.
Battery / paint / house cleaner / methanol / bicarbonate / plastic pack / pill capsule	supermarket & EMS buy lists	500 / 600 / 600 / 800 / 550 / 433 / 250	no buyback	—	KEEP	Buy-only precursors — correct. Never add buybacks.

* leaf-based costs value the grown leaf/oil at its own dealer price (what the player gives up by not selling it raw). New ladder check: leaves 356/402 → oil 862 → mdp2p 3,500 / methylamine 4,000 → mdma powder 5,500 → pill 6,200; powder 3,800 → pack 4,600. Every step goes up. ✓

D. Weapon dealer — every item

Military components come from the weapon-shop robbery (risk income → profit allowed). But the dealer must pay a **flat, similar rate per component** for every craftable, so no single item becomes the "loot ATM". Target: ~300–400 per component, guns highest.

Item	Source	Comps	Dealer pays now	/comp	NEW	new /comp	Action	Why
Normal/Reinforced barrel, Grip, Stock, Normal/Advanced trigger, Scope	items no longer exist	—	2,913–10,636	—	—	REMOVE	Ghost entries. If these items ever return, they are an instant exploit door.	
Knife	crafted (15 comps)	15	7,000	467	4,500	300	CHANGE	Was the 2nd-best loot ATM.
Machete	crafted (20 comps)	20	16,023	801	6,500	325	CHANGE	Worst hole in the dealer: old price was based on 4 reinforced steel, a recipe that no longer exists. 801/comp = spam machetes to cash out robbery loot.
Combat pistol	crafted (30 comps)	30	9,000	300	—	300	KEEP	On target.
Double-action revolver	crafted (45 comps)	45	16,506	367	—	367	KEEP	On target.
DB shotgun	crafted (60 comps)	60	22,086	368	—	368	KEEP	On target.
Micro SMG	crafted (65 comps)	65	20,506	315	—	315	KEEP	On target.
Sawn-off shotgun	crafted (70 comps)	70	27,667	395	—	395	KEEP	Highest rate = highest gang level. Good shape.
Compact rifle	crafted (90 comps)	90	28,638	318	—	318	KEEP	On target.
SMG / Pump shotgun / Advanced rifle / Combat pistol (shop-bought)	weapon shop 28,000 / 32,000 / 44,000 / 17,600	—	15,500 / 20,000 / 27,500 / 9,000	—	—	KEEP	Dealer pays ~50–62% of shop price → buy→sell always loses.	
Nightstick	police drop	—	4,000	—	—	KEEP	Risk item, small value.	

Vest (5 comps) / Lock pick (15 comps)	crafted	5 / 15	no buyback	—	—	KEEP	No buyback — keep it that way. Their value is use, not resale.
9mm ×200 / shells ×100 / 5.56 ×200	crafted (25 / 20 / 30 comps)	25/20/30	no buyback	—	—	KEEP	No buyback (Rule A7). Player-market ammo undercutting the shop (30/80/40 each) is intended gang income.

E. EMS / medical — every item (your "medic sells to players" design)

Buy prices stay. Sell-back prices drop **below crafting cost**, so the NPC circle always loses and the medic's real income is the player window. The unused `limitation` fields get deleted (rule A3).

Item	Materials (per unit)	Craft cost/unit	EMS shop pays now	NEW pays	Circle result	Player window	Action
Ethanol / Stabilizer / Pill capsule / Capsaicin / Acetone	EMS shop buy list: 800 / 250 / 250 / 1,000 / 2,000	—	no buyback	—	—	—	KEEP
Fabric	supermarket 200 (delete its dead buy-limit config)	—	no buyback	—	—	—	KEEP
Medical bandage	3 ethanol + 5 fabric → 5	680	900	550	−19% ✓	680 → 1,500 (shop buy)	CHANGE
Painkiller powder	4 stabilizer + 4 capsaicin → 5	1,000	no buyback	—	—	ingredient only	KEEP
Painkiller pill	5 capsule + 5 powder → 5	1,250	1,550	1,000	−20% ✓	1,250 → 2,000 (shop buy)	CHANGE
First-aid kit	4 bandage + 2 pill + 2 ethanol + 15 fabric → 2	4,910 (self-crafted)	6,500	4,400	−10% ✓	4,910 → 10,000 (shop buy)	CHANGE

Result: a medic who crafts a first-aid kit for ~4,910 sells it to players for e.g. 7,000–9,000 (cheaper than the 10,000 NPC shop → players happily buy). Selling to the NPC always loses ~10–20% → no bot loop, no cooldown needed. Exactly your design.

F. Junk dealer — every item

Item	Pays	Action	Why
Old circuit board / Bent pipe / Torn shoe / Crashed bottle / Rusty USB / Oil can / Broken phone / Damaged car battery / Broken lock pick	650 / 200 / 100 / 110 / 150 / 500 / 3,100 / 4,500 / 650	KEEP ALL	All come from searching bins (gathering time) and none can be bought from an NPC → no circle possible. Only thing to verify in code: bin search respawn/cooldown is server-side.

G. Food — burger job & shops

Item	Details	Now	NEW	Action	Why
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Ingredients (bun, meat, chicken, vegan meat, cheese, tomato, pickle, lettuce, onion, ketchup)	Food supplier, 7–50 each	—	—	KEEP	Buy-only, cheap, no buyback.
Sliced / cooked intermediates	benches	no buyback	—	KEEP	No NPC buys them → safe.
Hamburger (supermarket & vending price)	ingredients for 1 classic burger cost 257 ; NPC sells finished burger at 230	230	400	CHANGE	Broken player market (opposite direction): cooks can never compete with the NPC. At 400, cooks sell at ~300–350 — cheaper than NPC, above cost. Same as your medic design.
Double burger (supermarket price)	ingredients cost ~430; NPC sells at 420	420	650	CHANGE	Same reason.
Cooked food buyback	—	none	none	KEEP	Never add one (Rule A7) — cooks sell to players only.

H. Buy-only shops — confirmed safe (no sell-back exists)

Shop	Checked	Verdict
Supermarket, Vending machines, Booze shop, Gas station (fuel 4–8/L), Mechanic (fees + kits), Food supplier, Weapon shop (buy side), Clothes (normal + premium), Tattoo, Beauty salon, Plate shop, Insurance, DMV licenses, Dealerships + VIP, Rentals, Tuning, Garage	Every config file reviewed — none has a player-sells-to-NPC list	SAFE

Important: they are only safe after security fix A1 — today the generic sell handler would pay a hacked client for **any** item at **any** price, even items no shop buys.

I. Generator file changes — exact to-do

#	File	Change
1	CRAFTS/GANG.js	Replace all <code>gang_weapon_crafting</code> recipes with the military-component versions from the deployed config (knife 15, machete 20, lock pick 15, pistol 30, revolver 45, DB 60, micro SMG 65, sawn-off 70, compact 90, vests 5, ammo 25/20/30). Remove <code>craftTimeMinutes</code> everywhere.
2	JOBS/MINING.js	<code>IRON_ORE_BUY_PRICE 37 → 65</code> , <code>GOLD_ORE_BUY_PRICE 53 → 85</code> . Delete the commented-out blackSmith block (keep the empty array for the UI type).
3	SHOPS/DrugLab.js	methylamine 5700→ 3478 , mdp2p 5200→ 3043 , cocaine powder 4200→ 3304 , cocaine pack 5000→ 4600 , mdma powder 3900→ 4782 , mdma pill 9500→ 5391 (base values — ×1.15 multiplier gives the target prices from table C).
4	SHOPS/WeaponDealer.js	Delete the 7 ghost part entries; knife 7000→ 4500 ; machete 16023→ 6500 . Update the stale cost comments.

5	SHOPS/EMS.js	Sell list: bandage 900→ 550 , painkiller pill 1550→ 1000 , first-aid 6500→ 4400 . Delete <code>BuyLimitation</code> / <code>sellLimitation</code> and the <code>limitation</code> fields (dead).
6	SHOPS/SUPERMARKET.js + VENDINGMACHINE.js	Burger 230→ 400 ; double burger 420→ 650 (vending burger too).
7	generate_economy_config.js	Add the automatic circle-checker (fail build on any profitable all-NPC chain).
8	after generating	Copy <code>economyConfig.json</code> to both <code>high-ui/</code> and <code>high-roleplay-v1/src/src/shared/</code> . Never hand-edit the deployed JSON again.

J. Proof — every circle after the changes

Loop (all materials from NPC)	Material cost	NPC pays after fix	Result
100 iron ore + 10 coal → 4 iron bars	6,850	6,048	−12% ✓
40 iron + 100 gold ore + 10 coal → 4 gold bars	11,450	8,640	−25% ✓
3 methanol + 2 ammonia → methylamine	4,800	4,000	−17% ✓
1 mdma powder + capsule → pill (powder valued at its own new price)	5,750	6,200	+7.8% — allowed: powder needs grown leaves, not a pure NPC loop ✓
3 ethanol + 5 fabric → 5 bandages	3,400	2,750	−19% ✓
capsules + stabilizer + capsaicin → 5 painkiller pills	6,250	5,000	−20% ✓
bandages + pills + ethanol + fabric → 2 first-aid kits	9,820	8,800	−10% ✓
Shop weapon → weapon dealer	shop price	~50–62%	−38–50% ✓
Robbery components → machete/knife (risk income)	risk	300–395 /comp, flat	no "best item" ATM ✓

Order of work

1) Security fixes A1 + A2 (client price / client recipe) — without these, all prices are decoration. 2) Generator changes I1–I6, regenerate, copy to both consumers. 3) Circle-checker A5 so this stays fixed forever. 4) Delete dead timer/limit code A3 and add logging A4.

All costs computed from your real config files; component counts from the deployed server config; code paths verified in high-roleplay-v1 (SellService.ts, craftItem.ts).